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2.9 Plant Cell Structure

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What you will learn

- Special structures in plant cells and their role
- Structure and function of chloroplasts



[Figure 1]

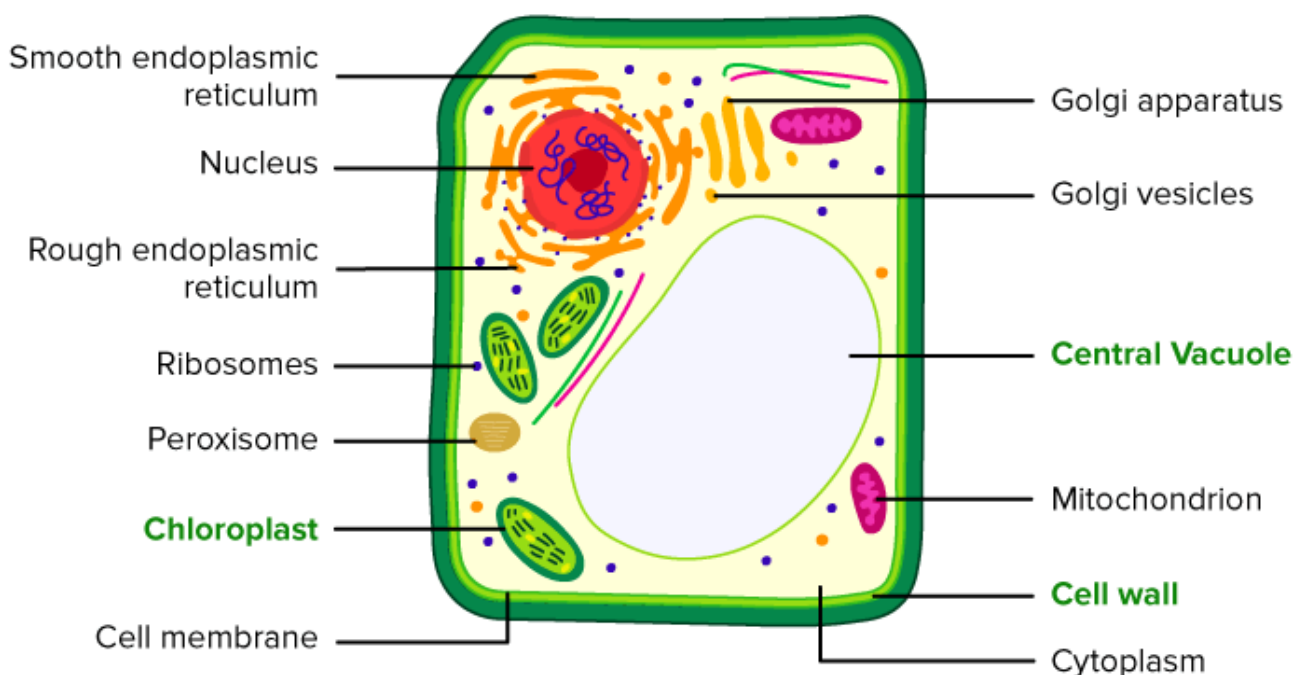
many plant cells are green. Why? Plant cells also usually have a distinct shape. The rigid exterior around the cells is necessary to allow the plants to grow upright. **Animal** cells do not have these rigid exteriors. There are other distinct differences between plant and animal cells. These will be the focus of this concept.

Plant Cells

Special Structures in Plant Cells

Most **organelles** are common to both animal and plant cells. However, plant cells also have features that animal cells do not have: a **cell wall**, a large **central vacuole**, and **plastids** such as chloroplasts.

Plants have very different lifestyles from animals, and these differences are apparent when you examine the structure of the **plant cell**. Plants make their own **food** in a process called **photosynthesis**. They take in carbon dioxide (CO₂) and **water** (H₂O) and convert them into sugars. The features unique to plant cells can be seen in the **Figure below**.



[Figure 2]

In addition to containing most of the organelles found in animal cells, plant cells also have a cell wall, a large central vacuole, and plastids. These three features are not found in animal cells.

The Cell Wall

A **cell wall** is a rigid layer that is found outside the **cell membrane** and surrounds the cell. The cell wall contains not only **cellulose** and **protein**, but other **polysaccharides** as well. The cell wall provides structural support and protection. Pores in the cell wall allow water and

Microtubules guide the formation of the plant cell wall. Cellulose is laid down by **enzymes** to form the primary cell wall. Some plants also have a secondary cell wall. The secondary wall contains a **lignin**, a secondary cell component in plant cells that have completed cell growth/expansion.

The Central Vacuole

Most mature plant cells have a **central vacuole** that occupies more than 30% of the cell's volume. The central vacuole can occupy as much as 90% of the volume of certain cells. The central vacuole is surrounded by a membrane called the **tonoplast**. The central vacuole has many functions. Aside from storage, the main role of the **vacuole** is to maintain turgor pressure against the cell wall. Proteins found in the tonoplast control the flow of water into and out of the vacuole. The central vacuole also stores the pigments that color **flowers**.

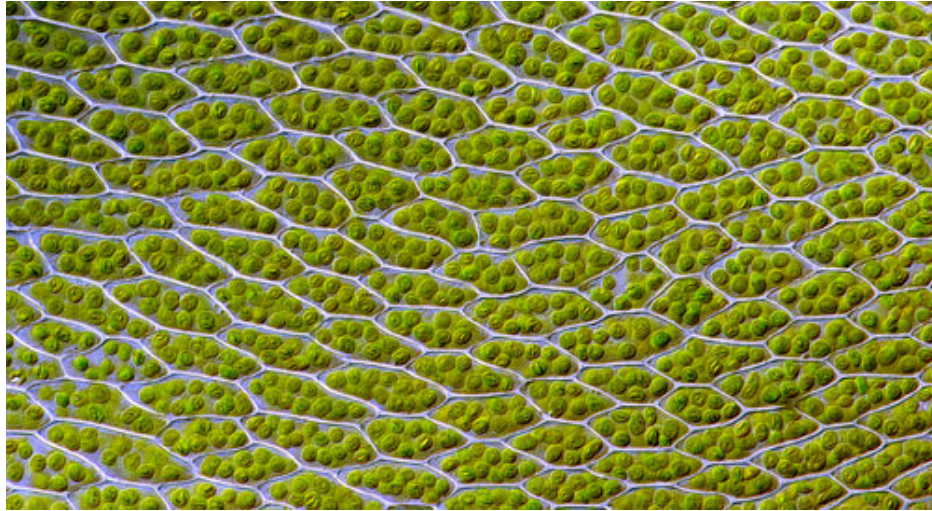
The central vacuole contains large amounts of a liquid called cell sap, which differs in composition to the cell **cytosol**. Cell sap is a mixture of water, enzymes, **ions**, salts, and other substances. Cell sap may also contain toxic byproducts that have been removed from the cytosol. Toxins in the vacuole may help to protect some plants from being eaten.

Plastids

Plant **plastids** are a group of closely related membrane-bound organelles that carry out many functions. They are responsible for photosynthesis, for storage of products such as **starch**, and for the synthesis of many types of molecules that are needed as cellular building blocks. Plastids have the ability to change their function between these and other forms. Plastids contain their own **DNA** and some **ribosomes**, and scientists think that plastids are descended from photosynthetic **bacteria** that allowed the first **eukaryotes** to make oxygen. The main types of plastids and their functions are:

- **Chloroplasts** are the organelle of photosynthesis. They capture light **energy** from the sun and use it with water and carbon dioxide to make food (sugar) for the plant. The arrangement of chloroplasts in a plant's cells can be seen in the **Figure below**.
- **Chromoplasts** make and store pigments that give **petals** and **fruit** their orange and yellow colors.
- **Leucoplasts** do not contain pigments and are located in **roots** and non-photosynthetic **tissues** of plants. They may become specialized for bulk storage of starch, **lipid**, or protein. However, in many cells, leucoplasts do not have a major storage function. Instead, they make molecules such as **fatty acids** and many **amino acids**.





[Figure 3]

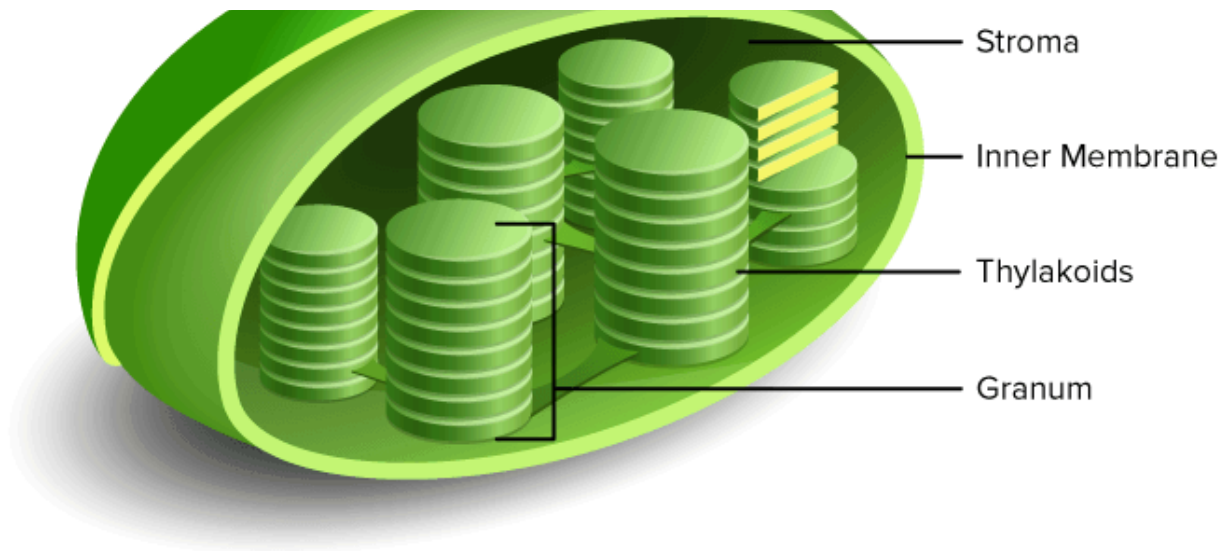
Plant cells with visible chloroplasts.

Chloroplasts

Chloroplasts capture light energy from the sun and use it with water and carbon dioxide to produce sugars for food. Chloroplasts look like flat discs and are usually 2 to 10 micrometers in diameter and 1 micrometer thick. A **model** of a chloroplast is shown in the **Figure below**. The chloroplast is enclosed by an inner and an outer phospholipid membrane. Between these two layers is the intermembrane space. The fluid within the chloroplast is called the **stroma**, and it contains one or more molecules of small, circular DNA. The stroma also has ribosomes. Within the stroma are stacks of **thylakoids**. The thylakoids are arranged in stacks called **grana** (singular: granum). A thylakoid has a flattened disk shape. Inside it is an empty area called the thylakoid space or lumen.

Within the **thylakoid membrane** is the complex of proteins and light-absorbing pigments, such as **chlorophyll** and carotenoids. This complex allows capture of light energy from many wavelengths because chlorophyll and carotenoids both absorb different wavelengths of light.





[Figure 4]

The internal structure of a chloroplast.

DID YOU KNOW?

Euglenoids are a unique group of unicellular **protists** with characteristics of both plant and animal cells. They have chloroplasts which help them make their own food in the presence of light. However, they can also live on nutrients in their surroundings in the absence of light.



Summary

- Plant cells have a cell wall, a large central vacuole, and plastids such as chloroplasts.
 - The cell wall is a rigid layer that is found outside the cell membrane and surrounds the cell, providing structural support and protection.
 - The central vacuole maintains turgor pressure against the cell wall.
 - Chloroplasts capture light energy from the sun and use it with water and carbon dioxide to produce sugars for food during photosynthesis.
-

Review

1. List three structures that are found in plant cells but not in animal cells.
2. Identify two functions of plastids in plant cells.
3. What are the roles of the cell wall and the central vacuole?
4. Describe the chloroplast structure.



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